



Pranav Khare <khare.pranav@gmail.com>

Invitation to Review Updated Article for IEEE Access Access-2024-34225

1 message

IEEE Access <onbehalf@manuscriptcentral.com>

Tue, Sep 10, 2024 at 4:49 PM

Reply-To: mafendee@uniswa.edu.my

To: khare.pranav@gmail.com

10-Sep-2024

Dear Mr. Khare:

Recently you reviewed an article submitted to IEEE Access entitled, "Beyond Classical Cryptography: A Systematic Review of Post-Quantum Hash-Based Signature Schemes, Security, and Optimizations". Please note that this article was rejected, and the authors were asked to update their article based on the reviewers' concerns.

The authors have resubmitted their revised article and supplied a "list of updates" that highlights all edits made, as well as their responses to the previous reviewers' concerns.

We would appreciate if you could take another look at this improved version to determine if your concerns have been addressed, and to ensure that you are satisfied with the updates.

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Once you accept my invitation to review this manuscript, you will be notified via email with directions on how to access the manuscript and the reviewer instructions in the Reviewer Center.

Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Dr. Mohamad Mohamed
IEEE Access Associate Editor
mafendee@uniswa.edu.my

MANUSCRIPT DETAILS

TYPE: Survey

TITLE: Beyond Classical Cryptography: A Systematic Review of Post-Quantum Hash-Based Signature Schemes, Security, and Optimizations

AUTHORS: FATHALLA, EFAT; Azab, Mohamed

ABSTRACT: The emergence of quantum computing poses significant risks to the security of current cryptographic systems, particularly those reliant on classical algorithms vulnerable to quantum attacks. This systematic literature review

adopts the PRISMA model to critically assess the development, methodologies, and security of post-quantum hash-based signature schemes as resilient alternatives. Through a methodical selection process from leading academic databases, we identify and analyze key contributions to the field within the last decade, focusing on the schemes' security proofs, enhanced performance, and efficiency metrics. Our analysis reveals a diverse landscape of hash-based signature schemes, their evolving security features against quantum threats, and their practical implementations in securing digital communications. The review highlights the importance of advancing these quantum-resistant technologies, discusses the challenges in their adoption, and outlines future directions for research and standardization efforts. The findings aim to provide a comprehensive resource for researchers, practitioners, and policymakers involved in the transition toward secure cryptographic practices in the quantum era.

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